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HoloGaitAR

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GitHub:

Introduction

Background

- HoloGaitAR is a Mixed reality gait rehabilitation system built upon the Microsoft HoloLens.
- Guides walking tasks using spatial waypoints and real-time cues.
- Captures session metrics and sends them to a web dashboard for progress review.

Rationale

- Inspired by the project owner's bicycle helmet-mounted metronome method.
- Existing approach provided rhythm cues but had limited feedback, adaptability and usability.
- Project modernizes this into a hands-free, spatial and data-driven interface.

Aims

- Develop a proof-of-concept to address limited supervision and inconsistent progress tracking.
- Improve accessibility for users with weaker motor control through simple interaction.
- Provide automatic performance logging to support clearer long-term monitoring.

Methodology

Project Initiation:

- **Problem Identification:** Began by analysing the existing helmet-metronome method and identifying gaps in accessibility.
- **Requirements Definition:** Functional goals were set for hands-free interaction, real-time guidance, automatic data capture.
- **Platform Familiarization:** Gained practical experience with HoloLens Development and AR/Mixed Reality workflows.

System Development & Integration:

- **System Architecture Design:** Designed pipeline linking HoloLens app, backend API, cloud storage, and web analytics dashboard.
- **Data Capture Pipeline:** Session Metrics were standardized into CSV for consistent logging and transfer.
- **Anomaly Detection Layer:** Statistical and clustering-based methods (z-score, k-means) were applied to highlight unusual gait sessions.
- **Accessibility Pivot:** The interaction design was shifted from voice-first controls to a gaze-based approach after user feedback showed that the project user could not reliably use voice commands.

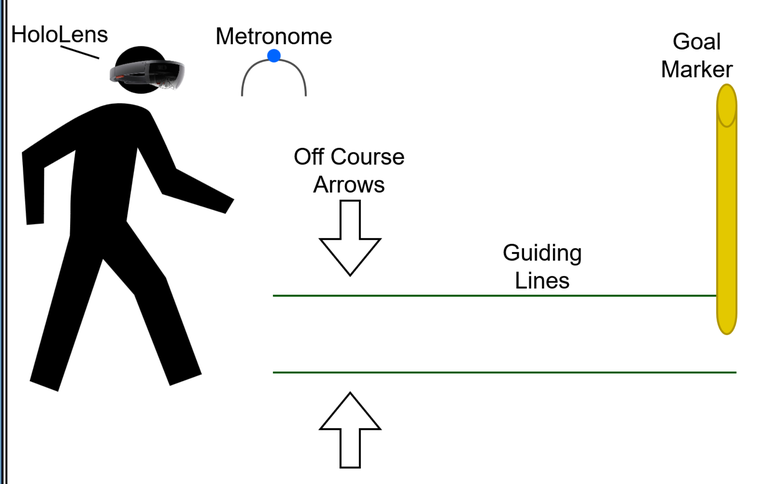
Validation & Refinement

- **Iterative Co-Design:** Ran repeated meetups with the project owner to validate features and refine the system based on direct feedback.

Conclusion

- **Clinical Fit depends on User Profile:** Although the HoloLens has practical limitations, the project owner feedback showed strong rehabilitation potential for users with milder impairments and upper-body control.
- **Accessibility Must Drive Interaction Design:** Real-world testing confirmed that control methods must be adapted to the user; shifting from voice-first to gaze-based interaction improved usability and inclusiveness without the cost of forfeiting features.
- **Objective Tracking Adds Clear Value:** Automatic session logging and dashboard analytics made progress more measurable and reviewable than manual cueing approaches.
- **Iterative Feedback is Essential:** Repeated testing & discussion with the project owner directly shaped design decisions and produced a more practical rehabilitation workflow.

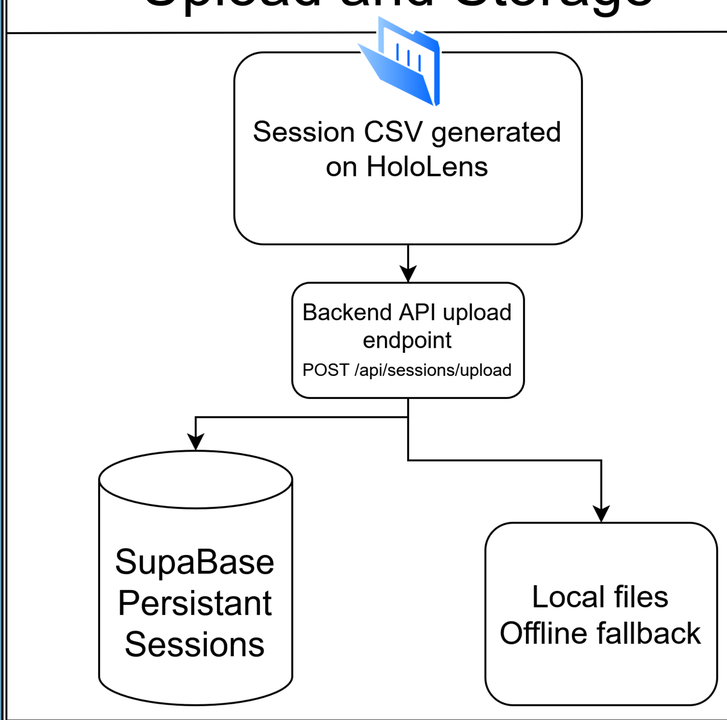
In-Session On HoloLens



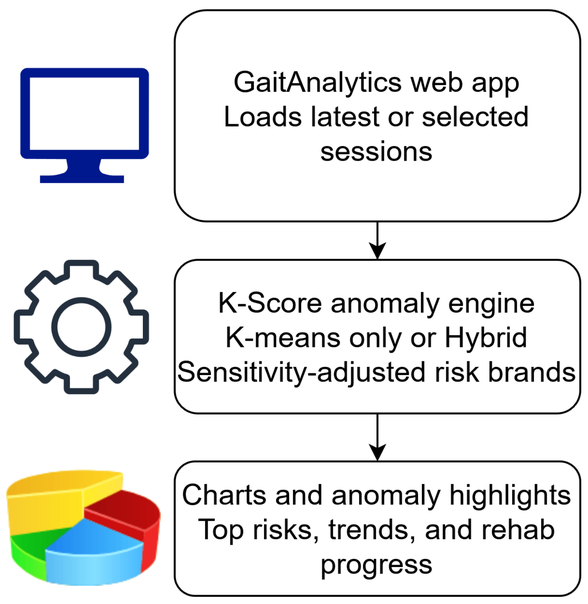
Captured Each Session

- Distance and elapsed time
- Speed and pace
- On-course and off-course percent
- Drift average and max

Upload and Storage



Web Dashboard Insight



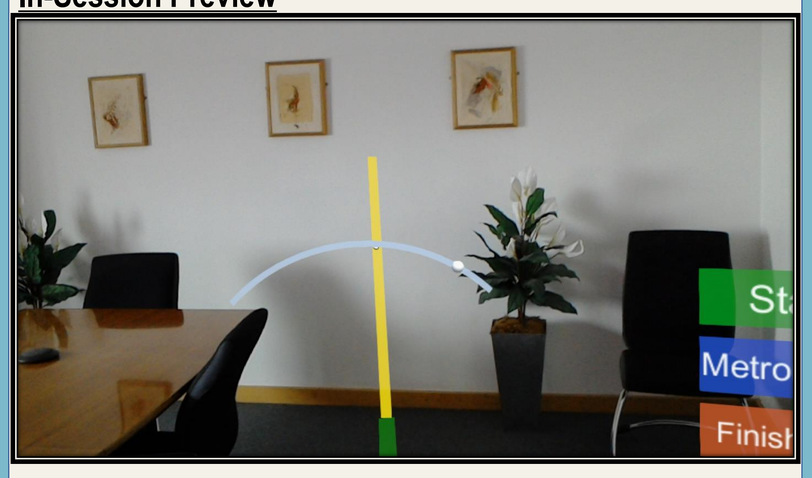
Feedback Loop: Insights shape progress and next rehab session

Limitations & Practical Fit

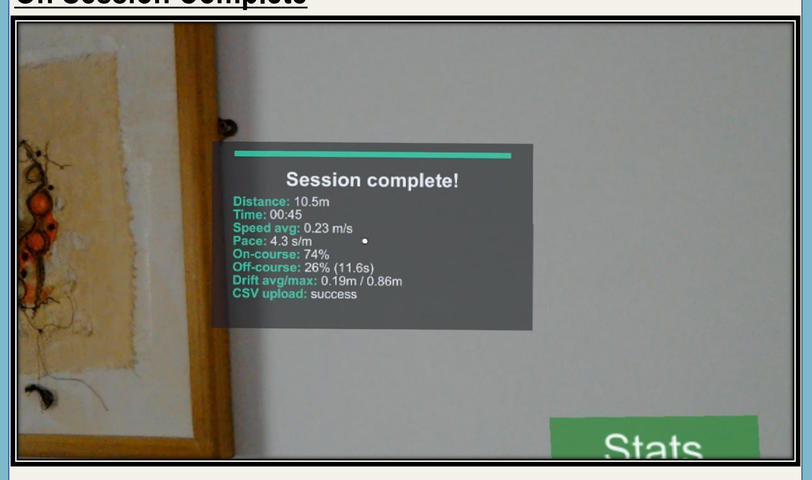
- HoloLens headset weight can reduce comfort during longer sessions.
- Limited field of view affects visibility of guidance elements.
- Interaction still depends on gaze and simple control actions, which may be difficult for users with severe impairments.
- Best current fit is users with mild to moderate motor impairment.

System Showcase

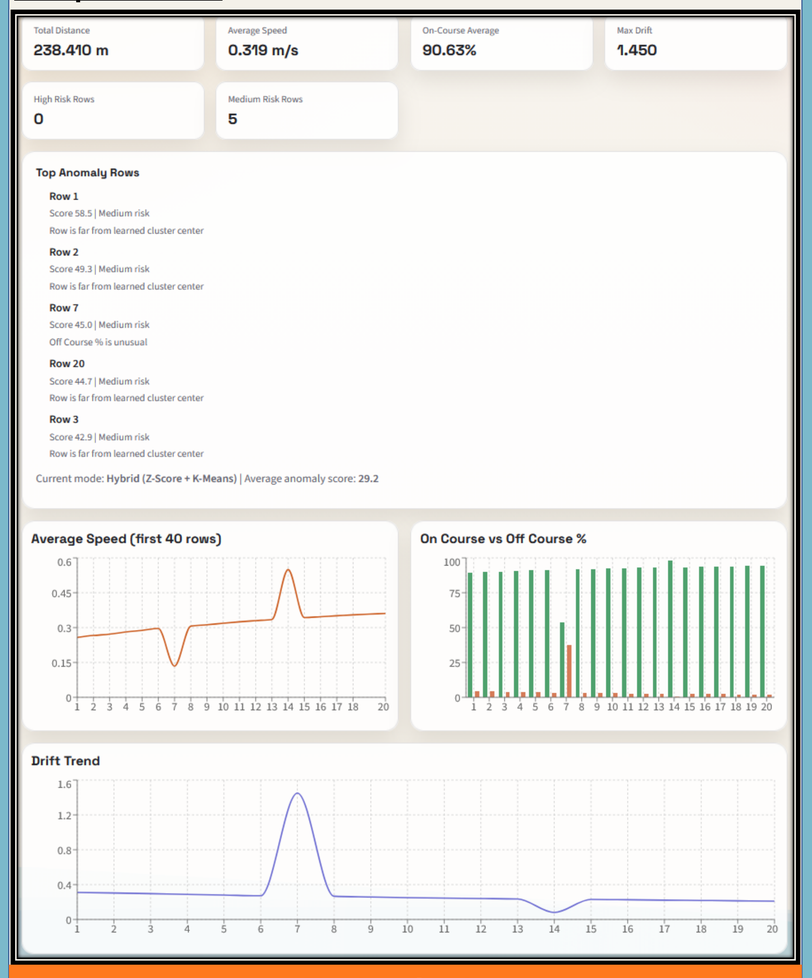
In-Session Preview



On Session Complete



Sample Metrics



Technologies

Software:

- **Unity (C#):** Core HoloLens rehabilitation app and session logic.
- **MRTK 2.8**: Mixed-reality UI, gaze, hand, and voice interaction support.
- **Node.js + Express:** Backend API for CSV upload and session retrieval.
- **Multer**: Handles CSV file uploads on the backend.
- **React + TypeScript:** Web dashboard for clinician-facing analytics.
- **Vite:** Fast frontend build and development tooling.
- **Recharts:** Data visualization for gait metrics and trends.
- **Papa Parse:** CSV parsing in the web dashboard.

Platform & Deployment:

- **Microsoft HoloLens:** Headset used for in-session gait guidance.
- **UWP:** Runtime/deployment target for the HoloLens app.
- **Supabase:** Persistent cloud database for session storage.
- **Render:** Cloud hosting for frontend and backend services.
- **GitHub:** Version control.